



Ou Ocean Kun Hei

HKUST(GZ)

BSc in Data Science and Big Data Technology
(Expected May 2027)

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EDUCATION

Hong Kong University of Science and Technology (Guangzhou)

2023 - Present

Relevant Coursework: Deep Learning, Introduction to Natural Language Processing and Knowledge Engineering, Database Management Systems, Machine Learning, Discrete Mathematics, Applied Statistics

CGPA: 3.362/4.4

PROJECT EXPERIENCE

DPC: Training-Free Text-to-SQL Candidate Selection via Dual-Paradigm Consistency (ACL2026)

X Program

2025.11 - 2026.1

- Addressing the issue that large language models exhibit "strong generation ability but weak selection ability" in Text-to-SQL tasks, we propose the DPC (Dual-Paradigm Consistency) framework, use a multi-agent collaborative approach with an adversarial feedback loop to synthesize Minimal Distinguishing Databases (MDD), and leverage the complementarity of SQL and Python in execution logic to construct model-independent prior reference anchors. As a plug-and-play module, it is integrated into state-of-the-art systems such as DAIL-SQL and CHESS, achieving an average accuracy improvement of 1.8%.
- As the second author, I was primarily responsible for conducting experiments across multiple models and systems, performing in-depth analysis of typical cases, and participating in paper revision and writing. The related work has been accepted by ACL 2026.

LinguaSQL: Unlocking NL2SQL Potential Through Multilingual and Multi-Dialect Prompting

X Program

2025.4 - 2025.12

- Developed a framework for generating candidate SQL queries with prompts in different natural languages and database dialects.
- Explored a selection strategy based on AST and gained approximately a 4% improvement in accuracy.

Large Model Self-Alignment Optimization System

Course Project

2025.12

Reproduced the core process of the self-alignment paper, constructed a reverse model of "answer → instruction" through LoRA fine-tuning to generate synthetic data. Forming a complete optimization closed loop from data synthesis and quality evaluation to model iteration.

Data Science Project

Course Project

2025.9 - 2025.12

Developed a personalized study-report feature in AnkiDroid open-source project, deploying a machine learning model via Flask. Implemented agile development and automated deployment using GitHub Workflow.

Campus Activities Registration and Management System

Course Project

2025.4 - 2025.5

Responsible for core backend development, independently completed the design and implementation of the entire data access layer and business logic layer, ensuring the efficiency and stability of key operations such as activity lifecycle management.

Exploring Power Outages in the Context of Climate Change and Energy Transition	Summer Project
Developed a climate-energy resilience model by analyzing global renewable energy and climate datasets in R, utilizing regression analysis to quantify future power outage risks under climate change scenarios.	2024.6 - 2024.9

SKILLS AND INTERESTS

Programming: Python, SQL, c++	
Frameworks: PyTorch	
Languages: English, Mandarin, Cantonese	
Areas of Interest: Data Engineering, Retrieval Augmented Generation, NL2SQL	

VOLUNTEER ACTIVITIES

PLACE Summer Camp, HKUST(GZ)	2024, 2025
15th National Games of the People’s Republic of China	2025.11

HONORS & AWARDS

Second-class Scholarship for Hong Kong, Macao, and Overseas Chinese Students	2025
Mirit Research Award of Undergraduate Research Program	2024
HKUST(GZ) Fisrt Class Scholarship	2023